

Regular Expression

Natural Language Processing

Ayesha Enayet

Reading Quiz

- Write the key takeaways from the paper (5–10 lines).

Regular Expression

- A language for specifying text search strings.
- An algebraic notation for characterizing a set of strings.
- A regular expression search function will search through the corpus, returning all texts that match the **pattern**.

What are Regular Expressions?

- Patterns for matching text; powerful for search, validation, and transformation.
- Typical uses: input validation, parsing logs, cleaning text, tokenization, extraction.
- In Python: use the standard library module ``re``.

Pattern

RE	Example Patterns Matched
/woodchucks/	“interesting links to <u>woodchucks</u> and lemurs”
/a/	“M <u>a</u> ry Ann stopped by Mona’s”
/!/	“You’ve left the burglar behind again <u>!</u> ” said Nori

Figure 2.1 Some simple regex searches.

RE	Match	Example Patterns
/[wW]oodchuck/	Woodchuck or woodchuck	“ <u>Woodchuck</u> ”
/[abc]/	‘a’, ‘b’, <i>or</i> ‘c’	“In uomini, in soldat <u>i</u> ”
/[1234567890]/	any digit	“plenty of <u>7</u> to 5”

Figure 2.2 The use of the brackets [] to specify a disjunction of characters.

RE	Match	Example Patterns Matched
/[A-Z]/	an upper case letter	“we should call it ‘ <u>D</u> renched Blossoms’ ”
/[a-z]/	a lower case letter	“ <u>m</u> y beans were impatient to be hoed!”
/[0-9]/	a single digit	“Chapter <u>1</u> : Down the Rabbit Hole”

Figure 2.3 The use of the brackets [] plus the dash - to specify a range.

RE	Match (single characters)	Example Patterns Matched
/[^A-Z]/	not an upper case letter	“O <u>y</u> fn pri <u>p</u> etchik”
/[^Ss]/	neither ‘S’ nor ‘s’	“ <u>I</u> have no exquisite reason for’t”
/[^\\.]/	not a period	“ <u>o</u> ur resident Djinn”
/[e^]/	either ‘e’ or ‘^’	“look up <u>^</u> now”
/a^b/	the pattern ‘a^b’	“look up <u>a^b</u> now”

Figure 2.4 The caret ^ for negation or just to mean ^. See below re: the backslash for escaping the period.

RE	Match	Example Matches
/beg.n/	any character between <i>beg</i> and <i>n</i>	<u>begin</u> , <u>beg'n</u> , <u>begun</u>

Figure 2.6 The use of the period . to specify any character.

Core Metacharacters (Cheat-Sheet)

- `.` → any char (except newline by default)
- `^` → start of string (or line with `re.MULTILINE`)
- `$` → end of string (or line with `re.MULTILINE`)
- `*` → 0 or more `+` → 1 or more `?` → 0 or 1
- `{m,n}` → between m and n repetitions
- `[...]` → character class; `[^...]` negates
- `(...)` → group (captures by default)
- `|` → alternation (logical OR)
- `\` → escape / special sequences (`\d`, `\w`, `\s`, etc.)

- `print(re.findall(r"\w+", "Hello 123!"))` # ['Hello', '123']
- `print(re.findall(r"\w+", "Hello 123!"))` # ['Hello', '123']
- `print(re.findall(r"\D+", "abc123xyz"))` # ['abc', 'xyz']

Note

- Please study from the reference book and the RegEx (python) documentation to learn more about Regular Expressions. It is part of your course, exam, quiz, and homework 😊

- `import re`
- `text = "Order #12345 was placed on 2025-08-28."`
- `print(re.findall(r"\d+", text))` # ['12345', '2025', '08', '28']

- `text = "A B\tC\nD"`
- `print(re.findall(r"\S", text))` # ['A', 'B', 'C', 'D']

Colab link for practice

- <https://colab.research.google.com/drive/1THp67kDpdnqEiPVSE85eqx-Y5Deyh21q?usp=sharing>

```
import re
def Date(string):
    DatePat = re.compile(r'\d{1,2}\d{1,2}\d{1,4}')
    string = DatePat.search(string)
    return string
print(Date("ABCD2/12/2024bbbbbbbbbb"))
```

<re.Match object; span=(4, 13), match='2/12/2024'>


```
import re

def Url(string):
    urlPat = re.compile(r'https?://(www\.)?[a-zA-Z0-9./]+')
    return urlPat.search(string)

print(Url("https://www.habib.edu University"))
# Output: <re.Match object; span=(0, 21), match='https://www.habib.edu'>
```

RegEx Module

Function	Description
<u>findall</u>	Returns a list containing all matches
<u>search</u>	Returns a <u>Match object</u> if there is a match anywhere in the string
<u>split</u>	Returns a list where the string has been split at each match
<u>sub</u>	Replaces one or many matches with a string

https://www.w3schools.com/python/python_regex.asp

https://www.w3schools.com/python/python_regex.asp

```
import re
```

```
#Check if the string starts with "The" and ends with "Spain":
```

```
txt = "The rain in Spain"
```

```
x = re.search("^The.*Spain$", txt)
```

```
if x:
```

```
    print("YES! We have a match!")
```

```
else:
```

```
    print("No match")
```

Output?

```
import re

txt = "The rain in Spain"
x = re.search("\s", txt)

print("The first white-space character is located in position:", x.start())
```

Output?

```
import re

txt = "The rain in Spain"
x = re.split("\s", txt, 1)
print(x)
```

```
import re
```

```
txt = "The rain in Spain"
```

```
x = re.sub("\s", "9", txt)
```

```
print(x)
```

Importance/Applications

- Data validation
 - Email validation, Password validation
- Search and Replace
- Data Extraction
 - URL extraction
- Data Cleaning